UDESH KUMAR

Principal Engineer · Frontend Platform Architecture · 16 Years

+91-9916118400 udeshvirk@gmail.com Bengaluru, India linkedin/udesh-kumar

PROFILE

Principal Engineer with 16+ years of experience architecting scalable frontend systems and driving platform-level engineering. At Zeta, I own the frontend platform powering 100+ applications across 20+ tenants and 50+ developers — covering project scaffolding, CI/CD automation, federated micro-frontend architecture, multi-tenant configuration, and S3-based artifact deployment.

I work at the intersection of platform engineering and developer experience — reducing delivery friction, standardising architecture, and enabling teams to ship faster through shared tooling, infrastructure, and a 50+ component design system.

EXPERIENCE

Principal Engineer — Frontend Platform

Nov 2021 – Present

Zeta Suite · Bengaluru

> *Own and evolve the platform infrastructure powering 100+ applications, 20+ tenants, and 50+ developers.*

- Designed and implemented a **federated dependency model** for micro-frontend architecture — enabling independent deployment of modules across **100+ apps** without cross-team coordination bottlenecks.
- Built a config-driven multi-tenancy system for **20+ tenants** — fully customising each tenant's environment (theming, feature flags, routing) through configuration rather than code changes.
- Designed an **extension framework** enabling runtime component extension — allowing applications to inject and override UI behaviours dynamically without redeployment.
- Built and maintain a **Kubernetes operator** that listens for frontend application resources and automates their deployment — removing manual infrastructure steps from the release process.
- Architected a platform service providing CLI scaffolding, shared build configs, and project templates — reducing new application setup from **days to hours** across 50+ developers.
- Owned end-to-end CI/CD pipelines — automating build, lint, test, and deployment workflows, eliminating manual release steps and standardising delivery across all teams.
- Built the artifact deployment pipeline: compiling bundles, publishing versioned assets to AWS S3, and configuring serving layers for zero-downtime deployments across **100+ applications**.
- Developed a centralised Design System of **50+ components** with design tokens, theming, and WCAG-compliant accessibility — adopted across all frontend teams.
- Led major platform refactoring — exception handling, error boundaries, and stability fixes — significantly improving platform reliability and developer confidence.
- Defined engineering standards: coding guidelines, test coverage requirements, and architectural decision records (ADRs) across **50+ developers**.

UI Lead

Nov 2017 – Oct 2021

ColorTokens, Inc. · Bengaluru

> *Led a team of 4–5 engineers building the admin console for a Zero Trust cybersecurity platform — agents on end-user devices block all traffic by default; admins define granular allow policies.*

- Led and mentored a team of **4–5 frontend engineers** — code reviews, technical direction, and engineering practice standards.
- Architected the frontend for a complex security admin console — policy editors, real-time traffic visualisations, and device management dashboards for enterprise clients.

- Designed and built a reusable component library and shared frameworks, improving consistency and reducing duplication across product areas.
- Optimised rendering performance for large policy datasets — improving responsiveness and reducing UI lag on complex views.
- Received merit recognition from clients and VP for outstanding delivery — including a merit-based salary hike.

Technical Lead — Development
Apr 2015 – Nov 2017

Relevance Lab Pvt. Ltd. · Bengaluru

- Led end-to-end delivery of web applications across multiple concurrent client engagements.
- Mentored engineers and established frontend coding standards, testing practices, and review processes.

Sr. Software Engineer
Mar 2014 – Mar 2015

Relevance Lab Pvt. Ltd. · Bengaluru

- Developed web application features and contributed to architecture and code quality initiatives.

EARLIER CAREER

Sr. Software Engineer · Baya Tree Pvt. Ltd. · Mohali*Oct 2012 – Feb 2014*

Sr. Software Engineer · Smart Data Enterprises · Mohali*Aug 2011 – Sep 2012*

Sr. Web Developer · RedAlkemi · Chandigarh*Jan 2009 – Aug 2011*

TECHNICAL SKILLS

FrameworksVue.js · React

Languages JavaScript (ES2020+) · TypeScript · HTML5 · CSS3

Platform Module Federation · Webpack · Vite · AWS S3 · CI/CD · Kubernetes

StylingSASS/LESS · Design Tokens

PracticesSDLC · Architecture Design · Performance Optimisation

EDUCATION

B.Tech – Computer Science & Engineering*2006 – 2009*

GGSCET, Punjab Technical University

Diploma – Information Technology *2006*

Government Polytechnic College, Punjab